Financial Time Series
46-929
Homework #2

Due: Thursday, April 8, 2020, 1:30pm.

1. **Weak Stationarity**

   a) (Differencing preserves weak stationarity)
   Suppose $\{X_t, \ -\infty < t < \infty\}$ is a weakly stationary process. Define a new process $Y_t = (1 - B)X_t = X_t - X_{t-1}$. Show that $\{Y_t, \ -\infty < t < \infty\}$ is also weakly stationary and find the mean, variance and autocovariance function of the Y_t process.

   b) (Linear combinations of independent weakly stationary processes are weakly stationary)
   Suppose $\{X_t^{(i)}, \ -\infty < t < \infty, \ i = 1, 2\}$ are weakly stationary independent. Show that $\{Y_t, \ -\infty < t < \infty\}$ where $Y_t = aX_t^{(1)} + bX_t^{(2)} + c$ is also weakly stationary.

2. **Introduction to the Yule Walker Equations**
   In lecture #3, we found that for an AR(2) process, $\{X_t, \ -\infty < t < \infty\}$ with $\{\epsilon_t, \ -\infty < t < \infty\}$ a WN$(0, \sigma_\epsilon)$ process, there was linear relationship between the model coefficients ϕ_1, ϕ_2 and the autocovariance function $\gamma(h)$ recalling that $\gamma(0) = \sigma_\epsilon^2$ and $\gamma(-h) = \gamma(h)$. This was used to find an iteration by which one could express $\gamma(h)$ or $\rho(h)$ as a function of ϕ_1 and ϕ_2 for any h.

   a) Turning this around, find two linear equations for ϕ_1, ϕ_2 in terms of ρ_1 and ρ_2. Express those two equations in matrix form. Solve to find a matrix expression for ϕ_1 and ϕ_2.

   b) Suppose we have estimates of $\gamma(1)$ and $\gamma(2)$, namely $\hat{\gamma}(1)$ and $\hat{\gamma}(2)$. One could plug them into your expression from part a) to obtain estimates for ϕ_1 and ϕ_2. Simulate 1,000 data points from an AR(2) sequence with iid $N(0, \sigma^2)$ errors and using those data, estimate ϕ_1 and ϕ_2. Use $\phi_0 = 0, \phi_1 = .33, \phi_2 = .33, \sigma_\epsilon^2 = .5$. Note, you can use the Python routine
   statsmodels.regression.linear_models.yule_walker(data, order =2, method = 'mle')
   as this 2x2 system is a very simple version of the general Yule-Walker equations to be covered in class.

   c) The accuracy of this method (assuming the data actually conforms to the model) depends upon σ_ϵ^2 and n, the size of the data. Repeat part b) with $n = 100$ and $n = 500$ to get an idea of how the error of estimation changes with the sample size.

3. **MA(1)**
   Consider a MA(1) process with mean 0,

$$X_t = \theta_1 \epsilon_{t-1} + \epsilon_t,$$

   for ϵ_t a WN$(0, \sigma^2)$. Pick your own coefficient θ_1 and let $n = 1000$. We worked out the autocovariance and autocorrelation function for some specific cases including MA(1). If one has a moment estimator for $\gamma(1)$, then one can use this to find a moment estimator for θ_1. Note, this can be done for higher-order MA processes, but the resulting equations are non-linear and difficult to solve. Simulate the MA(1) series of (length 1,000) 10 times with $\sigma^2 = 1$ each time using the simulated data to estimate θ. Report your 10 estimates and the variability of those estimates.

4. **Spurious Regression**

The problem of spurious regression (and spurious correlation) arises especially in economics where one performs a regression analysis with non-stationary time series data, and most economic time series are non-stationary. The simplest situation is where one has two non-stationary time series that are independent and regresses one on the other. Because the two series, while being independent, are both related to time, a significant but spurious correlation or regression can be found. Time is a confounding factor. The first report of this phenomenon dates back to G. Udny Yule in 1926, "Why do we Sometimes get Nonsense-Correlation between Time Series? – A Study in Sampling and the Nature of Time-Series." The most cited paper on this topic is the 1974 paper by Granger and Newbold, "Spurious Regressions in Econometrics,"(both papers are in the Course Papers folder). The simplest version of this phenomenon is where $\{X_t\}$ and $\{Y_t\}$ are independent random walks (which are non-stationary). That is $X_t = \sum_{i=1}^t Z_i^{(1)}$ and $Y_t = \sum_{i=1}^t Z_i^{(2)}$ where $\{(Z_i^{(1)}, Z_i^{(2)})\}$ are iid with independent $N(0, \sigma_j^2)$, $j = 1, 2$ distributions. Consider the model

$$Y_i = \beta_0 + \beta_1 X_i,\ 1 \leq i \leq t.$$

Clearly because the X and Y sequences are independent, $\beta_1 = 0$; however, one can still test $H_0 : \beta_1 = 0$ vs $H_1 : \beta_1 \neq 0$. Granger and Newbold did this 100 times with simulated data and found that roughly 75% of the trials resulted in rejection of H_0. In this problem let $\sigma_i^2 = 1,\ i = 1, 2$, simulate $\{(X_t, Y_t), 1 \leq t \leq 50)\}$ and test whether H_0 is significant at the 5% level. Do this 100 times and report the results.